Chapter 23

Electric Potential

Chapter Objectives

- Work done on a charged particle in an electric field
- The concept of electric potential
- Potential difference between one point and another

Chapter Outline

1. Electric Potential Energy
2. Electric Potential
3. Calculating Electric Potential
4. Equipotential Surfaces
5. Potential Gradient

23.1 Electric Potential Energy

- **Potential Energy** U is the work when a conservative force acts on a particle to move it from one point to another.

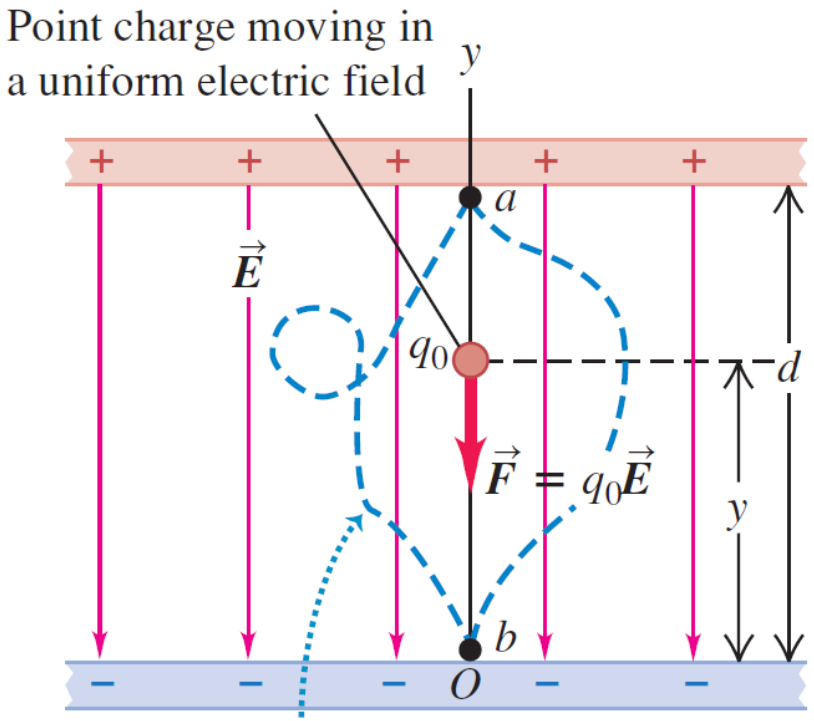
$$W_{a\to b} = U_a - U_b = -(U_b - U_a) = -\Delta U \qquad (23.2)$$

(work done by a conservative force)

Electric Potential Energy in a Uniform Field

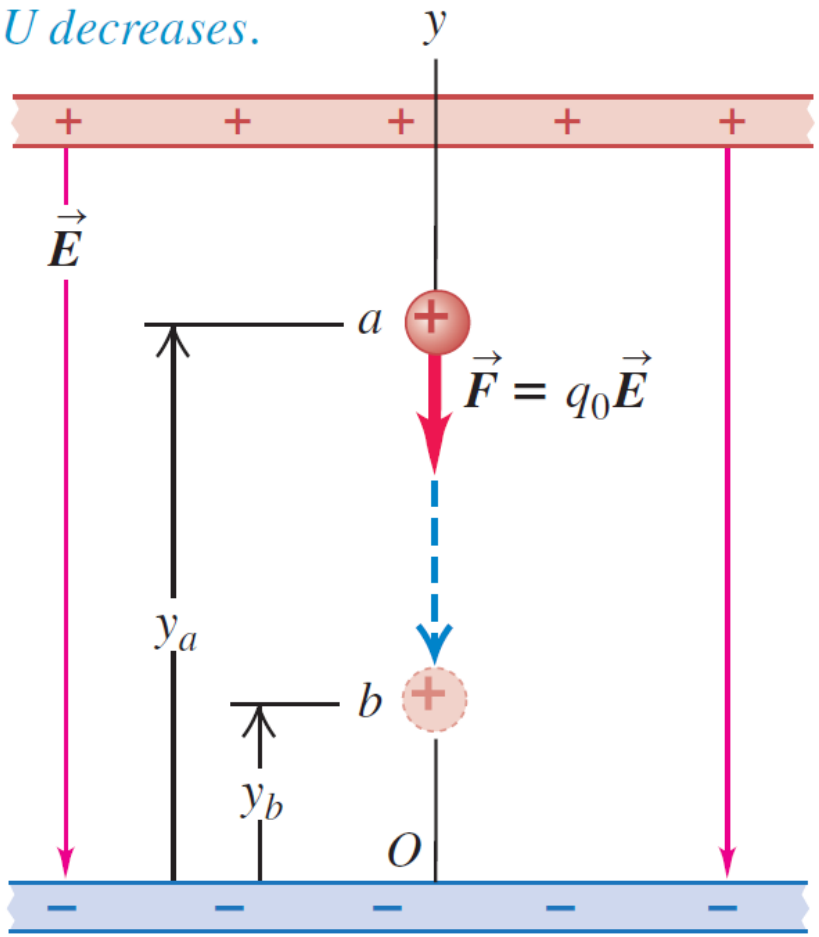
- The force on a test charge that moves from one point to another is constant and independent of its location.
- The work done by this force is also independent of the particle's path.

$$W_{a\to b} = -(U_b - U_a) = qE(y_a - y_b)$$



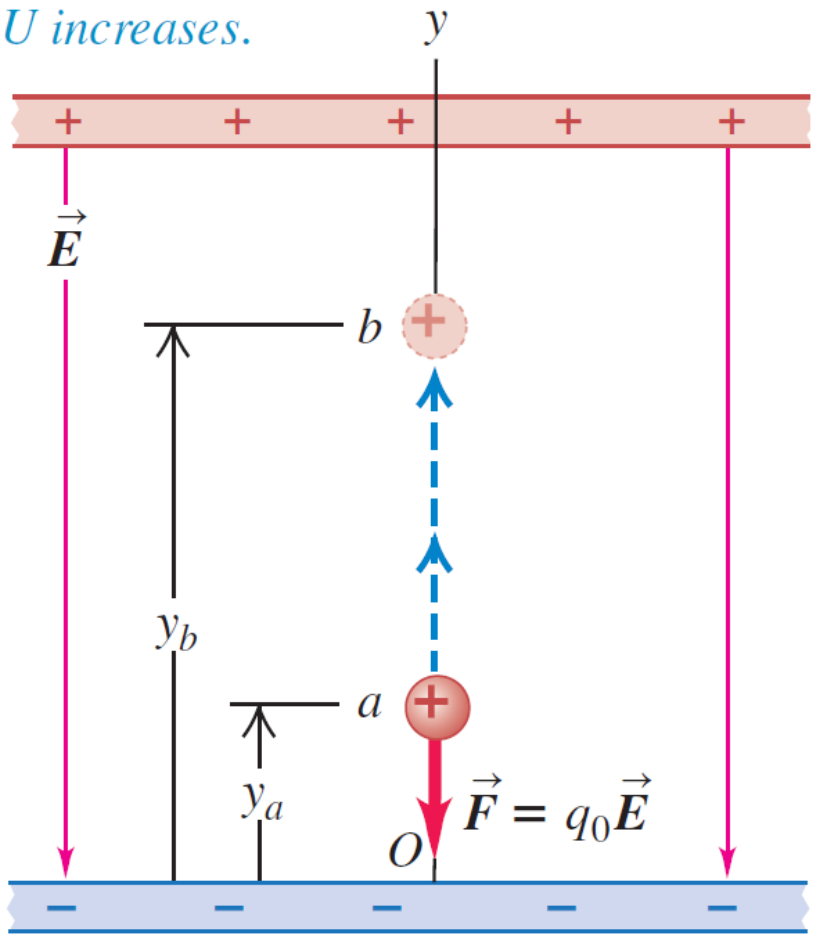
(a) Positive charge moves in the direction of $\vec{E}$:

- Field does *positive* work on charge.
- *U decreases.*



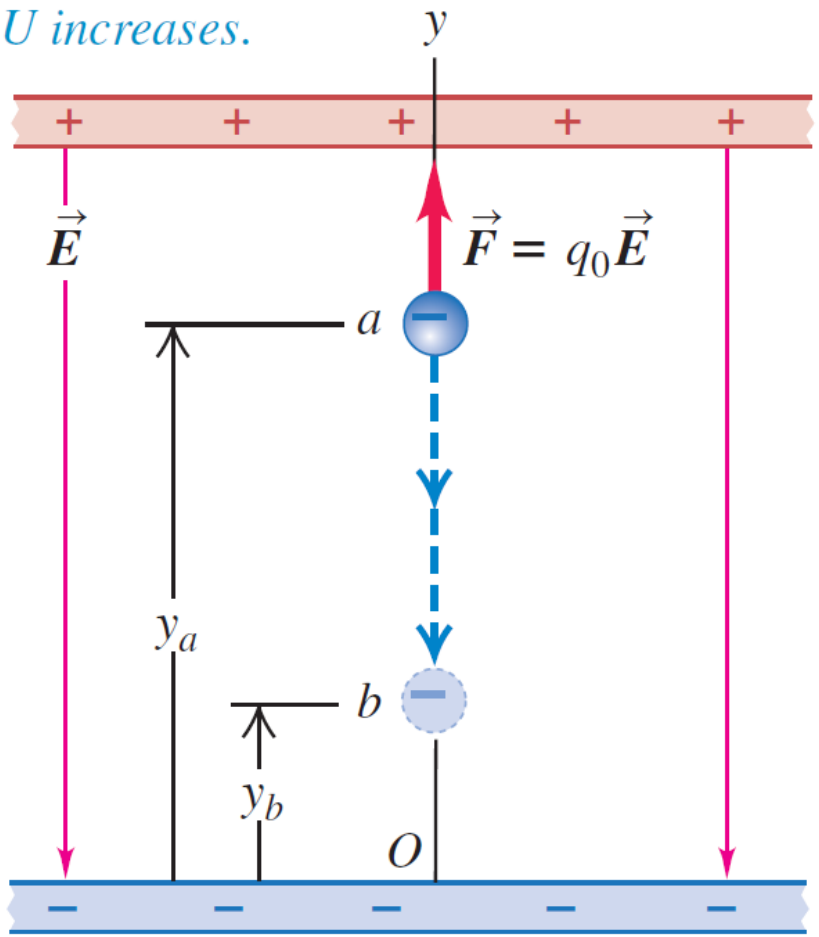
(b) Positive charge moves opposite $\vec{E}$:

- Field does *negative* work on charge.
- *U increases.*



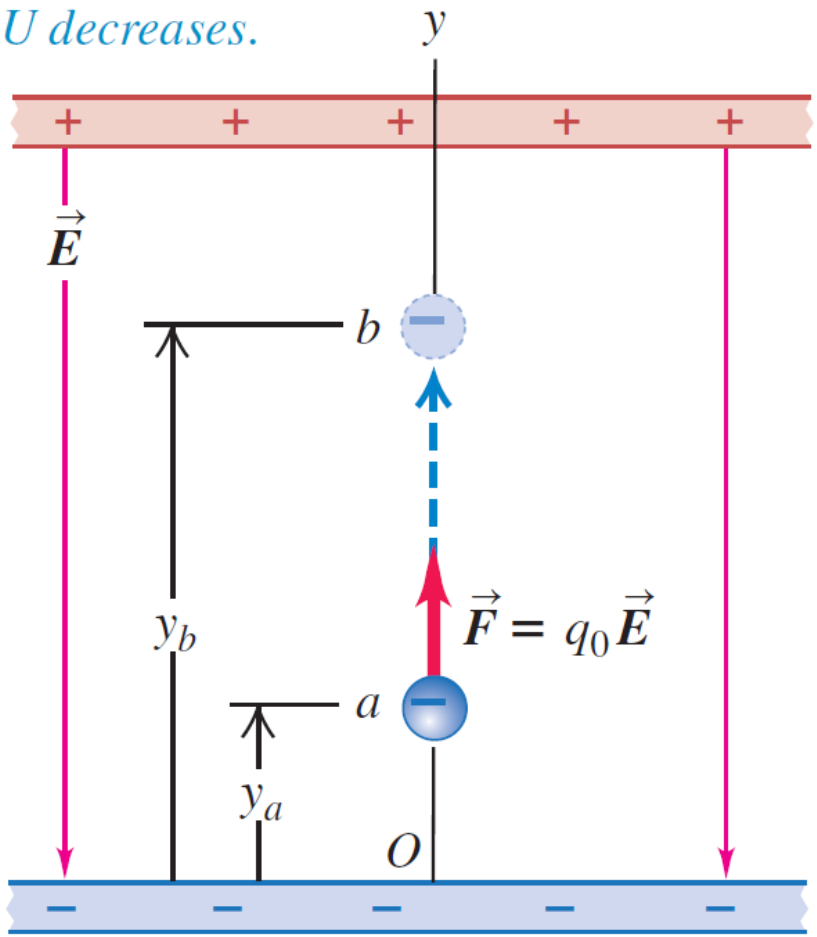
(a) Negative charge moves in the direction of $\vec{E}$:

- Field does *negative* work on charge.
- *U increases.*



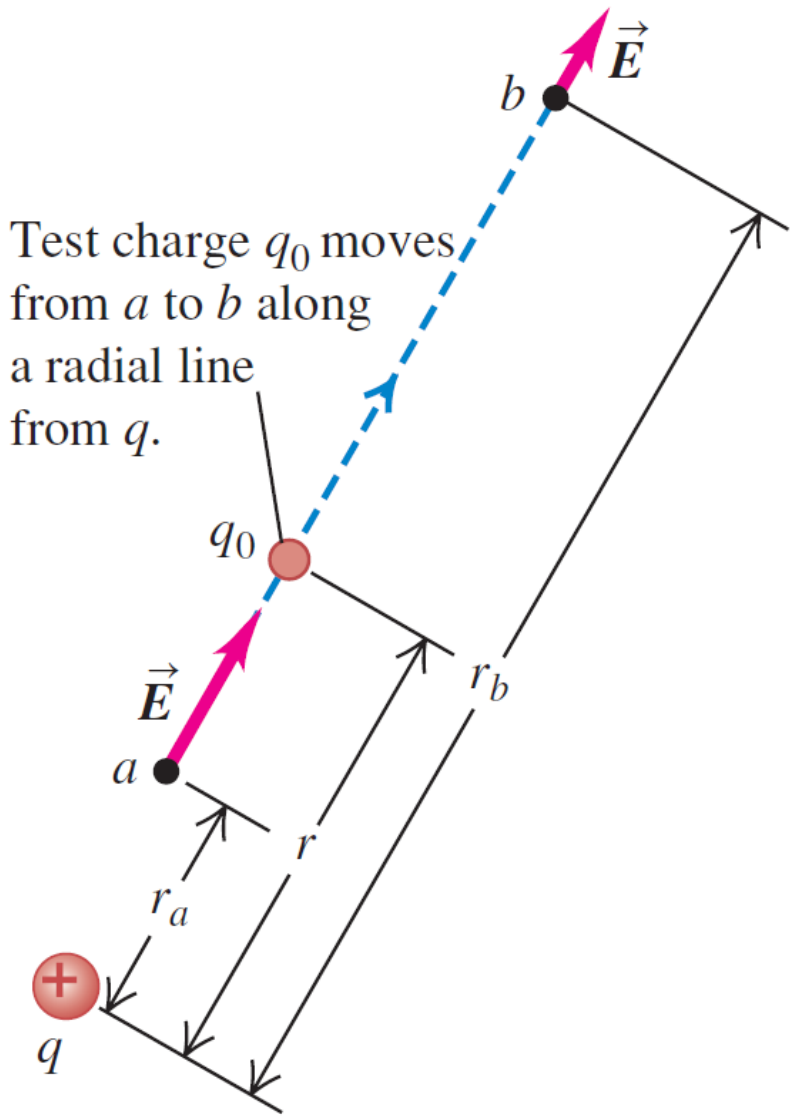
(b) Negative charge moves opposite $\vec{E}$:

- Field does *positive* work on charge.
- *U decreases.*



Electric Potential Energy in a Uniform Field

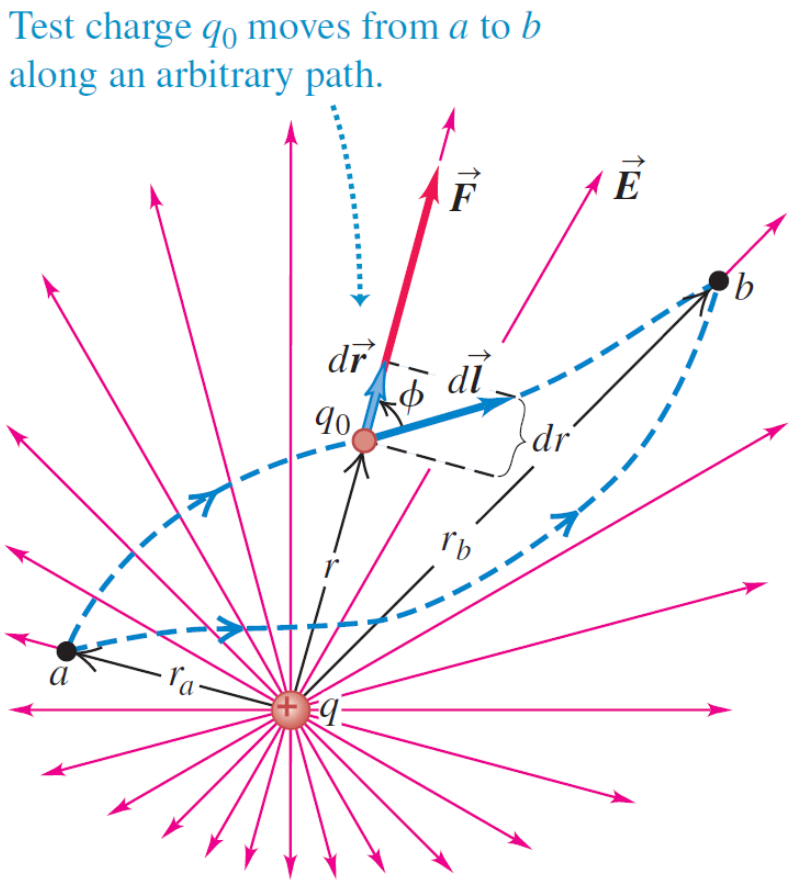
- The idea of electric potential energy isn't restricted to the special case of a uniform electric field.
- We can apply this concept to a point charge in any electric field caused by a static charge distribution.
- This is shown in the figure in the next slide, where displacement is along a radial line in the electric field set up by a single, stationary point charge.


Test charge q_0 moves from a to b along a radial line from q.
$\vec{E}$
b
q_0
$\vec{E}$
a
r_b
r
r_a
q

23.1 Electric Potential Energy

- In fact, the work done by the electric force in moving the test charge from points a to b is the same for all possible paths from a to b.

- This is shown in the figure on the right side where a and b do not lie on the same radial line and the test charge takes a non-linear path.



23.1 Electric Potential Energy

- The potential energy U when the test charge q_0 is at any distance r from charge q is thus

$$U = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r} \qquad (23.9)$$

(electric potential energy of two point charges q and q_0)

- Potential energy is always defined relative to some reference point where $U = 0$.

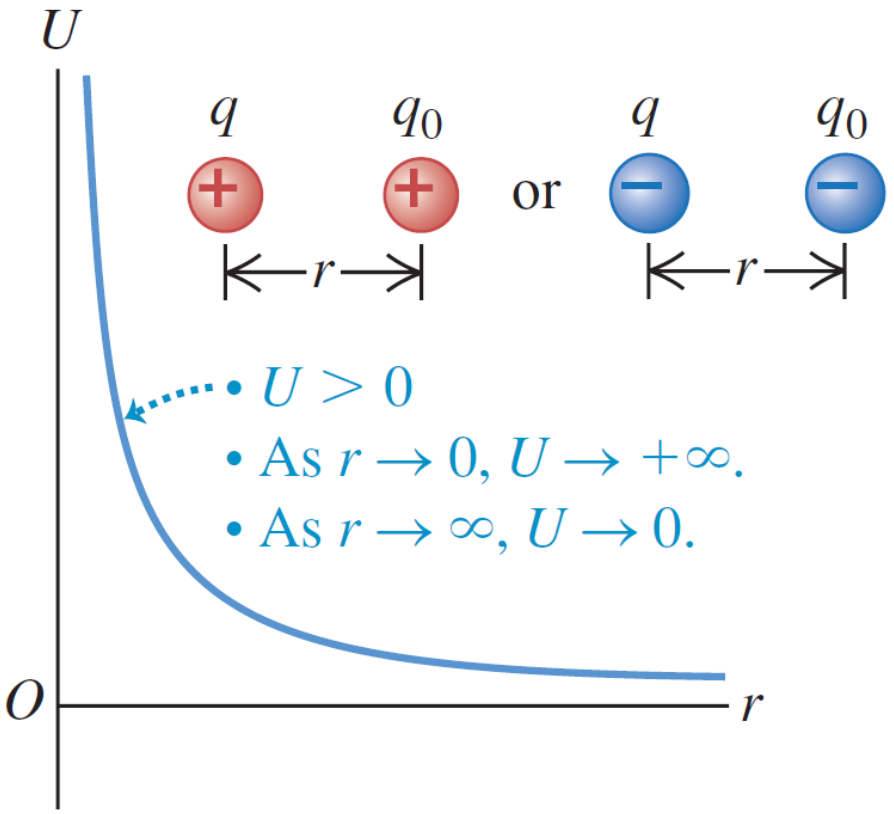
23.1 Electric Potential Energy

- It is a shared property of the two charges q and q_0 because it is a consequence of the interactions (attractive or repulsive) between these two bodies.

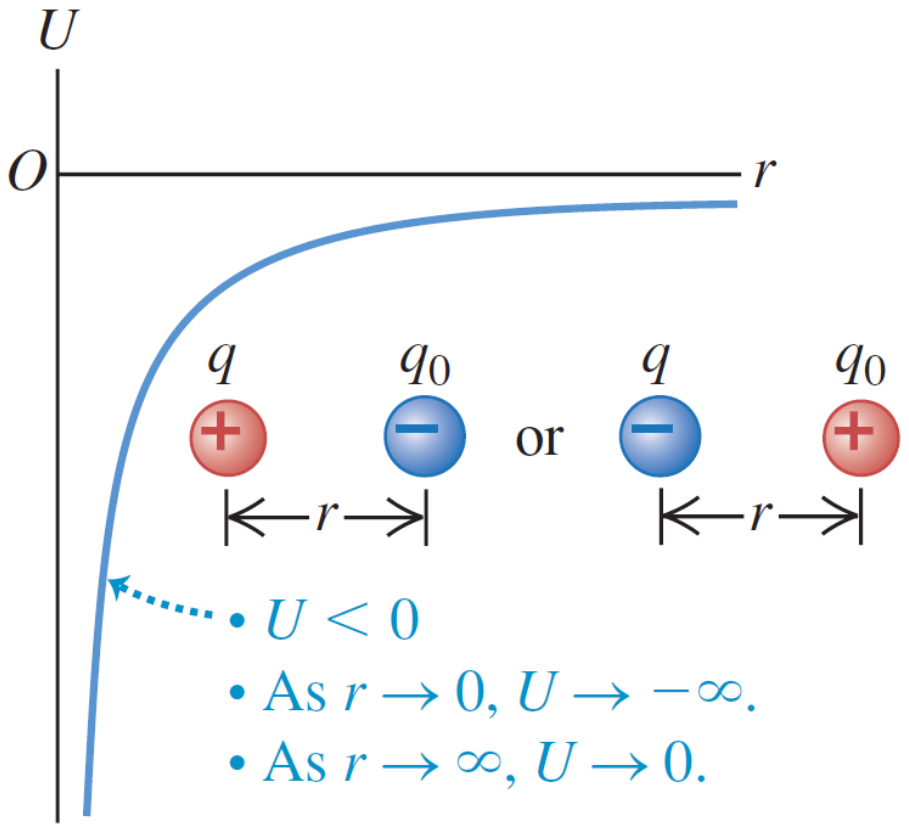
- Figure in the next slide is the graphical illustration of how U varies with these interactions.

23.1 Electric Potential Energy

(a) q and q_0 have the same sign.



(b) q and q_0 have opposite signs.



Example 23.1 Conservation of energy with electric forces

A positron (the antiparticle of the electron) has mass $9.11 \times 10^{-31}\ kg$ and charge $+e = +1.6 \times 10^{-19}\ C$. Suppose a positron moves in the vicinity of an alpha particle, which has charge $+2e = 3.2 \times 10^{-19}\ C$. The alpha particle is over 7000 times as massive as the positron, so we assume that it is at rest in some inertial frame of reference. When the positron is $1.00 \times 10^{-10}\ m$ from the alpha particle, it is moving directly away from the alpha particle at a speed of $3.00 \times 10^{6}\ m/s$.

Example 23.1 Conservation of energy with electric forces

a) What is the positron's speed when the two particles are $2.00 \times 10^{-10}\ m$ apart?

b) What is the positron's speed when it is very far away from the alpha particle?

c) How would the situation change if the moving particle were an electron (same mass as the positron but opposite charge)?

Identify

The electric force between the positron and alpha particle is conservative, so mechanical energy (kinetic plus potential) is conserved.

Set Up

The kinetic and potential energies at any two points a and b are related by $Eq.(23.3), K_a + U_a = K_b + U_b$, and the potential energy at any distance r is given by $Eq.(23.9)$.

We are given complete information about the system at a point a where the two charges are $1.00 \times 10^{-10}\, m$ apart. We use $Eq.(23.3)$ and (23.9) to find the speed at two different values of r in parts (a) and (b), and for the case where the charge $+e$ is replaced by $-e$ in part (c).

Execute

a) In this part, $r_b = 2.00 \times 10^{-10}\, m$ and we want to find the final speed v_b of the positron. This appears in the expression for the final kinetic energy, $K_b = \frac{1}{2} m {v_b}^2$; solving the energy-conservation equation for K_b, we have

$$K_b = K_a + U_a - U_b$$

The values of the energies on the right-hand side of this expression are

Execute

$$K_a = \frac{1}{2}mv_a{}^2$$
$$= \frac{1}{2}\left(9.11 \times 10^{31}\ kg\right)(3.00 \times 10^6\ m/s)^2$$
$$= 4.10 \times 10^{-18}\ J$$

$$U_a = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r_a}$$
$$= \left(9.0 \times 10^9\ N \cdot m^2/C^2\right)\frac{(3.20 \times 10^{-9}\ C)(1.60 \times 10^9\ C)}{1.00 \times 10^{-10}\ m}$$
$$= 4.61 \times 10^{-18}\ J$$

Execute

$$U_b = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r_b}$$
$$= \left(9.0 \times 10^9\ N \cdot m^2/C^2\right)\frac{(3.20 \times 10^{-9}\ C)(1.60 \times 10^9\ C)}{2.00 \times 10^{-10}\ m}$$
$$= 2.30 \times 10^{-18}\ J$$

Hence the final kinetic energy is

$$K_b = \frac{1}{2}m{v_b}^2$$
$$= 4.10 \times 10^{-18}\ J + 4.61 \times 10^{-18}\ J - 2.30 \times 10^{-18}\ J$$
$$= 6.41 \times 10^{-18}\ J$$

Execute

and the final speed of the positron is

$$v_b = \sqrt{\frac{2K_b}{m}}$$

$$= \sqrt{\frac{2(6.41 \times 10^{-18}\ J\ )}{9.11 \times 10^{31}\ kg}}$$

$$= 3.8 \times 10^6\ m/s$$

The force is repulsive, so the positron speeds up as it moves away from the stationary alpha particle.

Execute

b) When the final positions of the positron and alpha particle are very far apart, the separation r_b approaches infinity and the final potential energy U_b approaches zero. Then the final kinetic energy of the positron is

$$K_b = K_a + U_a - U_b$$
$$= 4.10 \times 10^{-18}\,J + 4.61 \times 10^{-18}\,J - 0$$
$$= 8.71 \times 10^{-18}\,J$$

Execute

And its final speed is

$$v_b = \sqrt{\frac{2K_b}{m}}$$

$$= \sqrt{\frac{2(8.71 \times 10^{-18}\ J\ )}{9.11 \times 10^{31}\ kg}}$$

$$= 4.4 \times 10^{6}\ m/s$$

Execute

Comparing to part (a), we see that as the positron moves from $r = 2.00 \times 10^{-10}\, m$ to infinity, the additional work done on it by the electric field of the alpha particle increases the speed by only about 16%.

This is because the electric force decreases rapidly with distance.

Execute

c) If the moving charge is negative, the force on it is attractive rather than repulsive, and we expect it to slow down rather than speed up.

The only difference in the above calculations is that both potential-energy quantities are negative.

From part (a), at a distance $r_b = 2.00 \times 10^{-10}\ m$ we have

Execute

$$K_b = K_a + U_a - U_b$$
$$= 4.10 \times 10^{-18}\ J + \left(-4.61 \times 10^{-18}\ J\right) - \left(-2.30 \times 10^{-18}\ J\right)$$
$$= 1.79 \times 10^{-18}\ J$$

$$v_b = \sqrt{\frac{2K_b}{m}} = 2.0 \times 10^6\ m/s$$

Execute

From part (b), at $r_b = \infty$, the kinetic energy of the electron would seem to be

$$\begin{aligned} K_b &= K_a + U_a - U_b \\ &= 4.10 \times 10^{-18}\, J + (-4.61 \times 10^{-18}\, J) - 0 \\ &= -5.1 \times 10^{-19}\, J \end{aligned}$$

But kinetic energies can never be negative!

This result means that the electron can never reach $r_b = \infty$;

Execute

The attractive force brings the electron to a halt at a finite distance from the alpha particle.

You can solve for the distance r_b at which the electron comes momentarily to rest by setting K_b equal to zero in the equation for conservation of mechanical energy (see Exercise 23.8).

Evaluate

It's useful to compare our calculations with Fig. 23.6.

In parts (a) and (b), the charges have the same sign; since $r_b > r_a$, the potential energy U_b is less than U_a.

In part (c), the charges have opposite signs; since $r_b > r_a$, the potential energy U_b is greater (that is, less negative) than U_a.

23.1 Electric Potential Energy

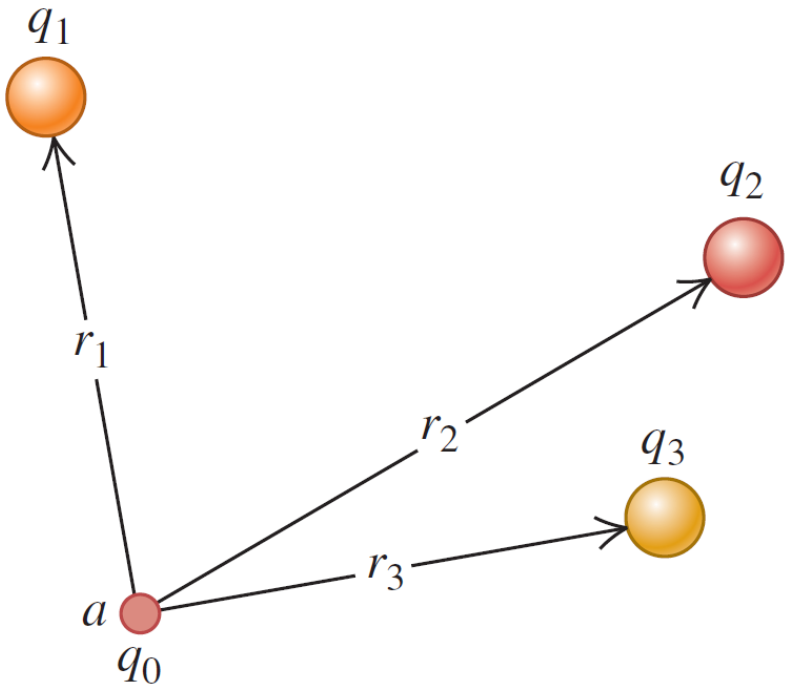
- The total electric field at each point is the vector sum of the fields due to the individual charges,
- And the total work done on the charge q_0 during any displacement is the sum of the contributions from the individual charges.

$$U = \frac{q_0}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \cdots\right) = \frac{q_0}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i} \qquad (23.10)$$

(point charge q_0 and collection of charge q_i)

23.1 Electric Potential Energy

- For every electric field due to a static charge distribution, the force exerted by that field is conservative.



Example 23.2 A system of point charges

Two point charges are located on the x-axis, $q_1 = -e$ at $x = 0$ and $q_2 = +e$ at $x = a$.

a) Find the work that must be done by an external force to bring a third point charge $q_3 = +e$ from infinity to $x = 2a$.

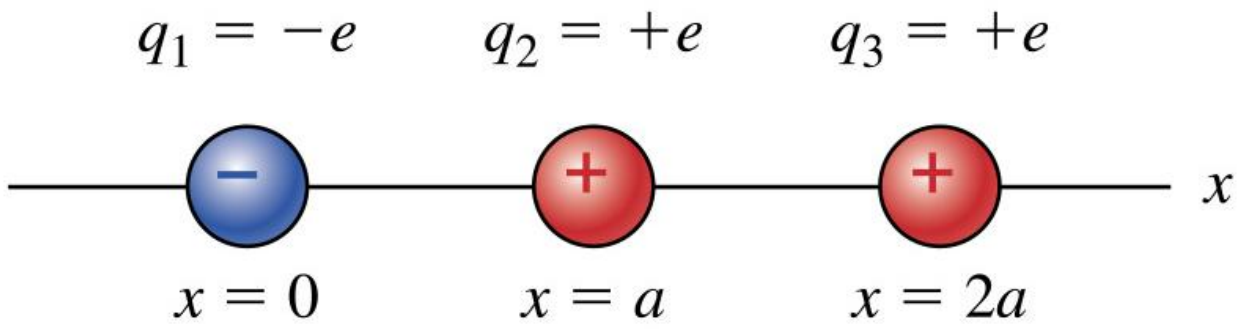
b) Find the total potential energy of the system of three charges.

Identify and Set Up

Figure below shows the final arrangement of the three charges.

To find the work required to bring q_3 in from infinity, we can find the potential energy associated with q_3 in the presence of q_1 and q_2.

We then can find the total potential energy of the system.



Execute

a) The work that must be done on q_3 by an external force $\vec{\boldsymbol{F}}_{ext}$ is equal to the difference between two quantities: the potential energy U associated with q_3 when it is at $x = 2a$ and the potential energy when it is infinitely far away. The second of these is zero, so the work that must be done is equal to U.

The distances between the charges are $r_{13} = 2a$ and $r_{23} = a$, so from Eq. (23.10),

Execute

$$W = U = \frac{q_3}{4\pi\varepsilon_0}\left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}}\right) = \frac{+e}{4\pi\epsilon_0}\left(\frac{-e}{2a} + \frac{+e}{a}\right) = \frac{+e^2}{8\pi\varepsilon_0 a}$$

If q_3 is brought in from infinity along the $+x$-axis, it is attracted by q_1 but is repelled more strongly by q_2; hence positive work done must be done to push q_3 to the position at $x = 2a$.

Execute

b) The total potential energy of the three charge system is

$$U = \frac{1}{4\pi\varepsilon_0}\sum_{i<j}\frac{q_i q_j}{r_{ij}}$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right)$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{(-e)(e)}{a} + \frac{(-e)(e)}{2a} + \frac{(e)(e)}{a}\right)$$

$$= \frac{-e^2}{8\pi\varepsilon_0 a}$$

Evaluate

Since our result in part (b) is negative, the system has lower potential energy than it would if the three charges were infinitely far apart.

An external force would have to do negative work to bring the three charges from infinity to assemble this entire arrangement and would have to do positive work to move the three charges back to infinity.